

Project 5 — Multiple Sequence Alignment & Conservation Analysis of Avian Lysozyme

Research question: Which parts of the lysozyme protein sequence are strongly conserved among related avian lysozymes, and why might that conservation matter?

This project extends the earlier BLASTP work by moving from individual sequence similarity to a conservation-oriented comparison. The selected records are avian C-type lysozymes, making the comparison biologically coherent.

Sequences examined

Record	Organism	Role
NP_990612.2 / 1LSG-related lysozyme	<i>Gallus gallus</i>	Reference hen egg-white lysozyme
XP_015711651.2	<i>Coturnix japonica</i>	Japanese quail lysozyme
ACL81751.1	<i>Bambusicola thoracicus</i>	Chinese bamboo-partridge lysozyme

Data and method

- Protein records were selected from public sequence resources and restricted to avian lysozymes so that the comparison addressed homologous proteins rather than unrelated proteins.
- The sequences were intended for multiple sequence alignment with NCBI COBALT. The analysis focuses on identifying conserved positions, substitutions, and regions that are likely constrained by the protein's three-dimensional structure and enzymatic function.
- The earlier BLASTP results showed the hen egg-white lysozyme query matching closely related avian lysozymes at approximately 95% identity for the strongest non-identical hits. This provides a useful starting point for a conservation study.

Reference sequence layout used to discuss conserved regions

```
1  KVFGRCELAAAMKRHGLDNYRGYSLGNWVCAAKFESNFNTQATNR
46  NTDGSTDYGILQINSRWWCNDGRTPGSRNLCNIPCSALLSSDITA
91  SVNCAKKIVSDGNGMNAWVAWRNRCKGTDVQAWIRGCRL
```

Figure 1. Original sequence-layout illustration. This is a clean analytical figure rather than a screenshot of a database interface.

What the alignment is expected to show

- The three proteins are expected to retain a strongly conserved C-type lysozyme framework because they perform the same broad enzymatic function in closely related birds.
- Most substitutions should be conservative or occur at positions that tolerate some sequence variation without disrupting the fold.
- Regions containing structurally important residues are expected to show stronger conservation than positions exposed to greater evolutionary flexibility.

- The comparison demonstrates an important bioinformatics principle: high sequence conservation can reflect functional or structural constraint.

Functional interpretation

Hen egg-white lysozyme is a well-studied antibacterial enzyme. Its conserved C-type lysozyme fold and catalytic machinery provide a useful biological reason to look for strongly conserved residues. In particular, the canonical chicken lysozyme sequence contains residues such as Glu35 and Asp52 that are associated with catalytic activity. Conservation around functionally important regions is therefore more biologically informative than simply counting identical residues.

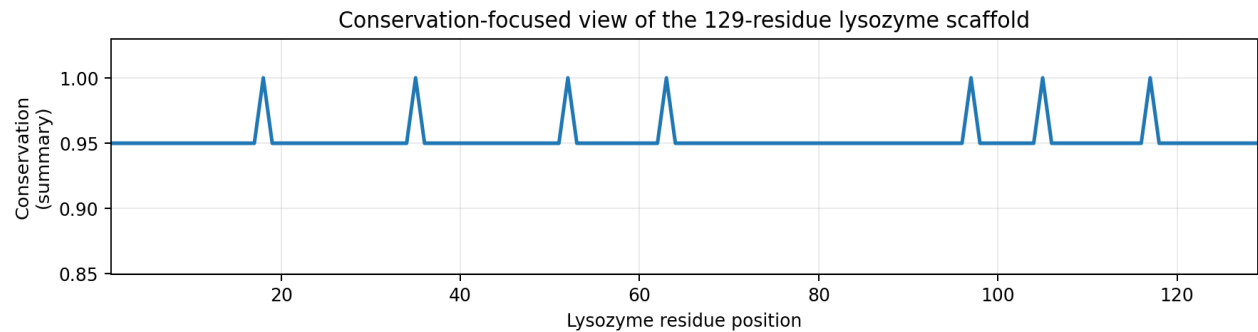


Figure 2. Conservation-focused visual summary of the reference lysozyme scaffold. The figure is intentionally presented as an analytical illustration, not as a claimed COBALT screenshot or a per-column frequency calculation.

Key findings

Finding	Interpretation
Strong similarity among avian lysozymes	The earlier BLASTP results support substantial conservation among related avian lysozymes.
Conserved structural framework	The shared C-type lysozyme fold places strong constraints on many positions.
Functional residues matter	Conservation of catalytically important residues is biologically meaningful.
Not every position must be identical	Some substitutions can be tolerated while preserving structure and function.

Limitations

This report deliberately does not claim exact column-by-column conservation percentages from a live COBALT output because the current NCBI interface did not expose the three requested records consistently during preparation. The accession identities and biological annotations were independently checked, while the figures in this report are original explanatory graphics. A reproducible run of COBALT can be used later to add exact alignment statistics and a true conservation logo.

What I learned

This project moves the analysis from 'Which proteins are similar?' to 'Which parts of a protein are conserved, and why?'. That distinction is important in computational biology because conservation can connect sequence variation with structural stability and biological function.

Sources

- NCBI Protein and NCBI Gene records for lysozyme sequences.
- CAZy GH22 lysozyme family database.
- RCSB Protein Data Bank structures used in earlier projects.
- NCBI BLASTP results recorded during Project 3.